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Lic. in Informatics and Computing Engineering (L.EIC)

Discrete Mathematics

4 - Semantics of First Order Logic

4.1. Consider the language $\mathcal{L}$, a language of first order with equality and such that $F_0 = \{a, b\}$, $F_1 = \{g\}$, $F_2 = \{f, h\}$, $R_1 = \{R, S\}$ and $R_2 = \{P, Q\}$. Let $\mathcal{A}$ be a structure of the language $\mathcal{L}$ defined by:

- the domain of $\mathcal{A}$ is the set of natural numbers $\mathbb{N} = \{0, 1, \ldots\}$
- $a^{\mathcal{A}} = 1$, $b^{\mathcal{A}} = 0$
- $g^{\mathcal{A}}(n) = n + 4$, $n \in \mathbb{N}$
- $h^{\mathcal{A}}(n, m) = n + m$, $f^{\mathcal{A}}(n, m) = n \times m$, $n, m \in \mathbb{N}$
- $S^{\mathcal{A}} = \{n \in \mathbb{N} \mid n \text{ is odd}\}$,$R^{\mathcal{A}} = \{n \in \mathbb{N} \mid n \text{ is even}\}$
- $P^{\mathcal{A}} = \{(n, m) \in \mathbb{N}^2 \mid n > m\}$
- $Q^{\mathcal{A}} = \emptyset$

If φ is each of the following propositions, determine whether it is true or false in $\mathcal{A}$, i.e. if $\mathcal{A} \models \varphi$ or $\mathcal{A} \not\models \varphi$.

(a) $\forall x \exists y\ P(y, x)$
(b) $\exists x \forall y\ P(y, x)$
(c) $\forall x \exists y\ (f(x, y) = a)$
(d) $\forall x \exists y\ P(y, x) \rightarrow \exists y\ P(y, y)$
(e) $\forall x\ S(x) \rightarrow \exists x\ S(x)$
(f) $\forall x\ (S(x) \vee R(x)) \rightarrow (\forall x\ S(x) \vee \forall x\ R(x))$
(g) $\forall x\ (S(g(x)) \rightarrow S(x))$

4.2. Consider the language $\mathcal{L}$, a language of first order logic without equality and such that $F_0 = \{a\}$, $F_1 = \{f\}$, $F_2 = \{g\}$, $R_1 = \{R\}$ and $R_2 = \{S\}$. For each of the following formulas of the following formulas ϕ, create two structures $\mathcal{A}$ and $\mathcal{B}$ such that $\mathcal{A} \models \phi$ and $\mathcal{B} \not\models \phi$:

a) $\forall x(\exists y(S(x, y) \rightarrow R(f(y))))$
b) $\exists z(R(f(z)) \wedge S(z, g(z, z)))$
c) $R(a) \rightarrow S(a, a)$
d) $\forall x(R(x)) \rightarrow \forall y(S(g(y, a), f(y)))$

4.3. Show that the following formulas are not semantically equivalent, indicating a language $\mathcal{L}$, formulas φ and ψ of $\mathcal{L}$, and a structure $\mathcal{A} = (A, \cdot^{\mathcal{A}})$ of $\mathcal{L}$, such that $\mathcal{A}$ is not a model of the corresponding formula.

a) $\forall x\,(\varphi \vee \psi)$ and $\forall x\,\varphi \vee \forall x\,\psi$

b) $\exists x\,\varphi \wedge \exists x\,\psi$ and $\exists x\,(\varphi \wedge \psi)$

c) $\forall x\,(\varphi \to \psi)$ and $\exists x\,\varphi \to \forall x\,\psi$

d) $\exists x\,(\varphi \to \psi)$ and $\exists x\,\varphi \to \exists x\,\psi$

e) $\forall x\,(\varphi \to \psi)$ and $\forall x\,\varphi \to \forall x\,\psi$

f) $\forall x \exists y\,\varphi$ and $\exists y \forall x\,\varphi$

4.4. For each of the following formulas determine an equivalent prenex normal form.

(a) $\neg\forall x\,R(x,u) \;\wedge\; \exists y\,(\neg\forall x\,R(y,x) \to Q(u,y))$

(b) $\forall x\,\neg\exists y \forall u\,P(x,y) \to \neg(\exists z\,P(x,y) \to \forall y\,P(u,y))$

(c) $\exists x\,S(x,x) \vee \exists z\,(P(z) \wedge \forall x\,R(x,z))$

Laws of semantic equivalence

$\phi \wedge \psi$	$\Leftrightarrow$	$\psi \wedge \phi$	(commutativity)
$\phi \vee \psi$	$\Leftrightarrow$	$\psi \vee \phi$	(commutativity)
$\neg(\phi \wedge \psi)$	$\Leftrightarrow$	$(\neg\phi \vee \neg\psi)$	(DeMorgan's Law)
$\neg(\phi \vee \psi)$	$\Leftrightarrow$	$(\neg\phi \wedge \neg\psi)$	(DeMorgan's Law)
$(\phi \wedge \psi) \wedge \theta$	$\Leftrightarrow$	$\phi \wedge (\psi \wedge \theta)$	(associativity)
$(\phi \vee \psi) \vee \theta$	$\Leftrightarrow$	$\phi \vee (\psi \vee \theta)$	(associativity)
$(\phi \vee \phi)$	$\Leftrightarrow$	ϕ	(idempotence)
$(\phi \wedge \phi)$	$\Leftrightarrow$	ϕ	(idempotence)
$(\phi \wedge \psi) \vee \theta$	$\Leftrightarrow$	$(\phi \vee \theta) \wedge (\psi \vee \theta)$	(distributiveness)
$(\phi \vee \psi) \wedge \theta$	$\Leftrightarrow$	$(\phi \wedge \theta) \vee (\psi \wedge \theta)$	(distributivity)
$\neg\neg\phi$	$\Leftrightarrow$	ϕ	(Double negation)
$\phi \to \psi$	$\Leftrightarrow$	$\neg\phi \vee \psi$	
$\neg\forall x\varphi$	$\Leftrightarrow$	$\exists x\neg\varphi$	(DeMorgan's Law)
$\neg\exists x\varphi$	$\Leftrightarrow$	$\forall x\neg\varphi$	(DeMorgan's Law)
$\forall x(\varphi \wedge \psi)$	$\Leftrightarrow$	$\forall x\varphi \wedge \forall x\psi$	(but not with $\vee$)
$\exists x\varphi \vee \exists x\psi$	$\Leftrightarrow$	$\exists x(\varphi \vee \psi)$	(but not with $\wedge$)
$\forall x \forall y\varphi$	$\Leftrightarrow$	$\forall y \forall x\varphi$	
$\exists x \exists y\varphi$	$\Leftrightarrow$	$\exists y \exists x\varphi$	
$\forall x(\varphi \wedge \psi)$	$\Leftrightarrow$	$\forall x\varphi \wedge \psi$	if x does not occur free in ψ
$\forall x(\varphi \vee \psi)$	$\Leftrightarrow$	$\forall x\varphi \vee \psi$	if x does not occur free in ψ
$\exists x(\varphi \wedge \psi)$	$\Leftrightarrow$	$\exists x\varphi \wedge \psi$	if x does not occur free in ψ
$\exists x(\varphi \vee \psi)$	$\Leftrightarrow$	$\exists x\varphi \vee \psi$	if x does not occur free in ψ